Finding a Hidden Planet in Noisy Starlight

A Smart pipeline that detects and classifies exoplanet transits in TESS data • ISRO / PRL Hackathon — Problem Statement 7 • Project Report

1. The Problem

When a planet passes in front of its star, it blocks a tiny sliver of the star's light — the star appears to dim by a fraction of a percent for a few hours, and this repeats every orbit. NASA's TESS space telescope records the brightness of thousands of stars so we can hunt for these dips. The trouble is that the dips are **buried in noise**: detector static, the star's own flickering, and light leaking in from neighbouring stars in crowded parts of the sky all hide the signal. Worse, not every dip is a planet — two stars orbiting each other (an 'eclipsing binary'), a faraway pair blended into the same pixel, or big star-spots can all fake the same dimming. Checking 20,000–30,000 stars per sector by hand is impossible.

> **The goal:**
> Build a program that automatically (1) finds the dips, (2) decides whether each one is a real planet or a look-alike, (3) says how confident it is, and (4) measures the planet's orbit and size — all with no human hand-holding.

2. What Our Project Does

We built **VyomNetra**, an end-to-end pipeline that takes a raw, noisy TESS light curve and returns a labelled, confidence-scored planet candidate together with a clear diagnostic picture. It runs on any target just by giving it a star's catalogue ID, downloads real data live from NASA's archive, and needs no answers typed in by hand. It also handles the tricky case of a planet that transits only *once* in the data by looking at the shape of that single dip — exactly as our PRL mentors advised.

3. How It Works

Step 1 — Get the data. Download the star's brightness measurements (its 'light curve') straight from the public TESS/MAST archive using SAP_FLUX, as recommended by the mentors.

Step 2 — Clean it up. Remove slow drifts and glitches so real dips stand out, while being careful not to erase the transit itself. This cut the noise on our test star roughly in half.

Step 3 — Hunt for a repeating dip. Slide a search across thousands of possible orbit lengths to find a regular dimming (the Box-Least-Squares method). A second detector separately catches lone, one-off dips.

Step 4 — Fold and stack. Line up every orbit on top of the others so the tiny repeated dip adds up and becomes obvious — this is called phase-folding.

Step 5 — Run the lie-detector tests. Three physical checks rule out the common fakes (binary stars, blends, grazing stars) and score how planet-like the signal is.

Step 6 — Measure & classify. Look up the star from the catalogue, work out the planet's size, orbit distance and temperature, attach a confidence score, and print the label.

4. Our Result

We ran the full pipeline on **TIC 35516889**, which the archive lookup identifies as **WASP-19** — host to WASP-19 b, one of the fastest-orbiting 'hot Jupiters' known, whose entire year lasts under 19 hours. Starting from raw, noisy flux and with no values supplied by us, the pipeline produced the picture below and correctly flagged a **planet candidate** with very high significance (signal ~200× above the noise).

5. Our Accuracy:

We compared each pipeline number against the accepted values in the NASA Exoplanet Archive. The orbit and distance are essentially spot-on; the depth and duration come out a little small, for reasons we explain below.

What we measured	Our pipeline	NASA archive	Match
Orbital period	0.78863 days	~0.78884 days	**99.97%**
Distance from star (a)	0.01654 AU	~0.0165 AU	**99.7%**
Equilibrium temperature	2038 K	~2100 K	**97%**
Transit duration	1.44 hours	~1.57 hours	**92%**
Planet radius	1.29 R_Jupiter	~1.41 R_Jupiter	**92%**
Transit depth	16,657 ppm	~20,000 ppm	**83%**

Why are depth and duration a bit low?

- **Light from neighbours.** WASP-19 has faint companion stars nearby. Their extra glow leaks into the same pixels and ‘fills in’ part of the dip, making it look shallower. Professional teams spend weeks removing this; we ran on raw data in seconds.
- **Our fast search rounds the corners.** The Box-Least-Squares method fits a simple rectangular ‘box’ to a naturally curved dip, so it clips the gentle edges and reports a slightly shorter, shallower event. A slower curve-fitting step (on our to-do list) recovers those few minutes.

6. How Do We Know It’s a Real Planet, Not a Fake?

A deep, regular dip is not enough — we have to rule out the impostors. The pipeline runs three physical checks, and WASP-19 b passed all three cleanly:

All three checks passed → genuine planet candidate.

✓ **Check 1: Do the dips take turns?** If two stars were eclipsing each other, every second dip would be a different depth. Ours match almost perfectly (difference of 0.01σ) — so it is one object crossing, not a pair of stars.

✓ **Check 2: Is there a hidden second dip?** A glowing companion star would show a smaller dip halfway between the main ones. We found only a tiny 400 ppm blip — far too small for a star, exactly what a planet gives.

✓ **Check 3: Is the dip flat-bottomed?** A planet fully crossing the star gives a flat-bottomed 'U' shape; a grazing pair of stars gives a pointy 'V'. Ours is 76% flat — a clean U-shape.

Combining the three checks with the raw signal strength, the pipeline reports an overall **confidence of 91.6%**, with the transit standing a remarkable **~200 times** above the background noise (well beyond the standard detection bar of 7×).

7. Tools We Used

Everything is free and open-source Python: **Lightkurve** and **Astroquery/MAST** to fetch real TESS data and star catalogues; **Astropy** for the Box-Least-Squares transit search; **NumPy** and **SciPy** for cleaning and the single-dip detector; **scikit-learn** for the (upcoming) classifier and PR-AUC scoring; and **Matplotlib** for the diagnostic picture. Assumptions we made: the raw SAP_FLUX fairly represents the brightness, one sector of data is enough for screening, and a simple albedo value is used for the temperature estimate.

8. Conclusion

From raw, noisy TESS data — with no human hints and nothing hard-coded — our pipeline rediscovered WASP-19 b, matched its 19-hour orbit to 99.97%, ruled out every common impostor, and labelled it a genuine planet candidate at 91.6% confidence. Finishing the machine-learning classifier and the blend test will turn this student prototype into a survey-grade planet-hunting tool.